

Scientific Reporting (IMRaD)

Lecture: write section-by-section with clarity, integrity, and strong visuals

Dr Kanthu Joseph Mhango

2026-01-22

Goal of this lecture

This is a **scientific reporting lecture** focused on writing and presenting **IMRaD**:

- **Introduction** (why did you start?)
- **Methods** (what did you do?)
- **Results** (what answer did you get?)
- **Discussion** (what does it mean?)

We will not cover “the scientific method” basics.

Session overview

- Why IMRaD exists + how to read the structure as a “contract with the reader”
 - **Introduction** — question, gap, aim, contribution, and framing
 - **Methods** — experimental design + measurement + analysis transparency
 - **Results** — descriptive + inferential, table design, and visualisation (deep dive)
 - **Discussion** — interpretation, limits, implications, and claims discipline
-

Why a rigid structure?

Scientific writing is a **creative process** (rhetoric + science), but journals enforce structure because it:

- helps authors remember **what to include** and **in what order**
 - helps reviewers/editors evaluate completeness
 - helps readers locate information without reading everything
-

IMRaD

From the historical argument for structure, readers need authors to answer:

- **Why did you start?** → **Introduction**
 - **What did you do?** → **Materials & Methods**
 - **What answer did you get?** → **Results**
 - **What does it mean?** → **Discussion**
-

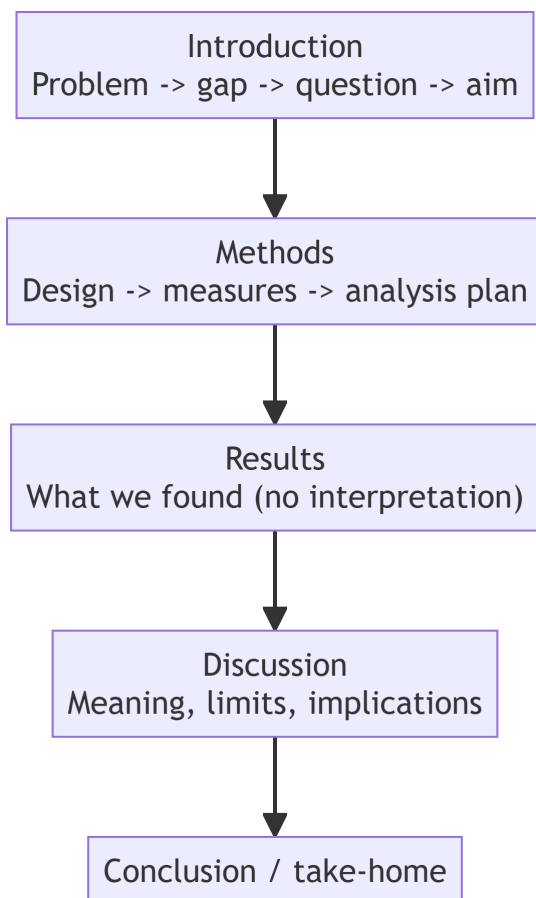
What IMRaD buys you (as a reader)

- **Facts vs interpretation:**
 - facts: **Methods + Results**
 - comment/theory: **Introduction + Discussion**
 - **Modular reading** (especially digitally): jump to what you need
 - **Findability:** easier to locate key information
 - **Less clutter:** discourages unnecessary detail
-

Before you write: align to the journal + reporting guidelines

- Always check the journal's **Instructions to Authors**
 - Use the right reporting guideline for your study type:
 - **CONSORT** (randomised trials)
 - **STROBE** (observational studies)
 - **PRISMA** (systematic reviews/meta-analyses)
 - **STARD** (diagnostic accuracy)
-

IMRaD as a “reader journey”



Writing principle (applies everywhere)

Respect the contract:

- **Intro:** motivates + frames the question
 - **Methods:** makes evaluation and replication possible
 - **Results:** presents the evidence cleanly
 - **Discussion:** interprets without over-claiming
-

The hidden triad: claims, evidence, and uncertainty

Every paper implicitly argues:

- **Claim:** what you want the reader to believe
- **Evidence:** what you measured/analysed
- **Uncertainty:** how wrong you might be

Strong IMRaD keeps these aligned and explicit.

Common failure mode

When papers are hard to trust, it's usually because:

- the **question** is unclear (Intro)
 - the **design/measurement** is underspecified (Methods)
 - the **results** are over-compressed or selectively shown (Results)
 - the **claims exceed the evidence** (Discussion)
-

Roadmap

Next we go section-by-section:

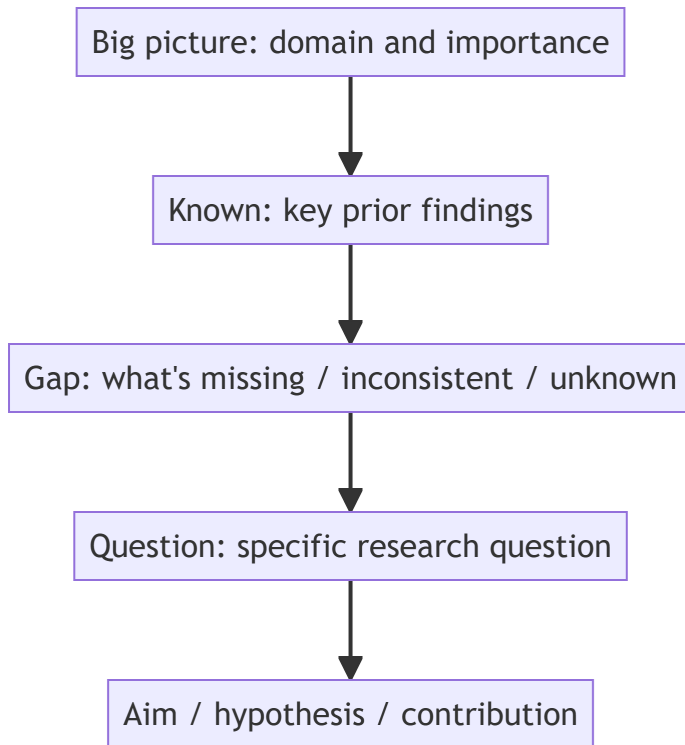
- **Introduction:** the problem funnel
 - **Methods:** design principles + reproducibility
 - **Results:** descriptive → inferential → tables → figures
 - **Discussion:** disciplined interpretation
-

Introduction (the “why did you start?” section)

The Introduction must answer:

- What problem exists?
 - What is already known?
 - What is the gap/limitation?
 - What is your **research question** and **aim**?
 - Why should the reader care **now**?
-

Introduction structure: the funnel



Intro: build a “problem statement” paragraph

One paragraph that includes:

- **Context** (1–2 sentences)
 - **Gap** (“However...”, “Yet...”, “Little is known about...”)
 - **Consequence** (why the gap matters)
 - **Your move** (“Therefore, we...”)
-

Intro: define your terms

Define only what the paper *needs*:

- ambiguous key terms
- population and context boundaries
- how you'll use terms consistently

Avoid: - long history lessons - definitions that do not matter to the design or interpretation

Intro: write the research question as a testable sentence

Use a one-sentence question with explicit variables:

- Population/context
 - Exposure/intervention or predictor(s)
 - Outcome(s)
 - Comparator (if applicable)
 - Time window (if relevant)
-

Intro: hypotheses and objectives

Be explicit about:

- **Primary objective** (the one the study is powered for / centred on)
- **Secondary objectives** (supporting or exploratory)
- **Hypothesis** (directional when justified)

Avoid “hypothesis fishing” in the Intro: don't pretend everything was planned.

Intro: “paragraph about statistics used”

Many journals expect the Intro to mention (briefly):

- what type of evidence you will present (e.g., group comparisons, regression)
- the primary outcome and planned comparison/model

Keep it high-level; details belong in Methods.

Intro: the last paragraph

End your Introduction with:

- your aim + question
 - study design (in 5–10 words)
 - primary outcome(s)
 - one sentence preview of the paper structure (optional)
-

Intro: quick quality checklist

- Does the reader know *exactly* what you tested?
 - Is the gap real (supported by citations)?
 - Is your contribution clear (new data, new method, new synthesis)?
 - Can someone predict what will appear in Methods/Results?
-

Methods (the “what did you do?” section)

Methods exist so readers can judge:

- Was the question approached **thoroughly**?
 - Were the tools/design **appropriate**?
 - Could the work be **replicated**?
-

Methods: rule of thumb

Write Methods as if the reader wants to answer:

- “Could I **repeat this** exactly?”
 - “Could I **audit this** for bias and errors?”
-

Methods: standard subheadings

- Study design and setting
 - Participants / sampling / eligibility criteria
 - Variables, measures, instruments
 - Interventions/exposures (if any)
 - Procedure and timeline
 - Outcomes (primary/secondary)
 - Sample size / power (when applicable)
 - Statistical analysis plan
 - Ethics approvals / consent (if applicable)
 - Data/code availability (where appropriate)
-

Methods: experimental design principles

Even in non-lab domains, these ideas transfer:

- **Control/comparison:** what is the baseline?
 - **Randomisation:** reduces selection bias and confounding
 - **Blinding:** reduces measurement and expectation bias
 - **Replication:** stabilises estimates and reveals variability
 - **Standardisation:** same protocol across units/time
-

Methods: define the experimental unit

One of the most common errors:

- confusing **experimental unit** (what was randomised) with
- measurement unit (what was measured)

This affects:

- independence assumptions
 - valid sample size (n)
 - appropriate modelling (clustered / repeated measures)
-

Methods: controls and comparators

Specify:

- what the control group/condition represents
 - how comparability was ensured (randomisation, matching, adjustment)
-

Methods: randomisation

If randomised:

- method (simple, block, stratified)
 - How random allocation was assured
-

Methods: replication and repeated measures

Be explicit about:

- number of replicates per condition
 - whether replicates are:
 - biological: different plants/plots etc
 - technical: same plant, measured multiple times to estimate measurement error
 - temporal: same plant, replicaion over time
 - how repeated measures were handled (averaging vs modelling)
-

Methods: inclusion/exclusion criteria

Report eligibility criteria so readers can judge:

- selection bias risk
 - generalisability limits
 - whether exclusion introduces collider bias (in observational studies)
-

Methods: measurement (variables + units + instruments)

For each key variable:

- definition (operationalisation)
 - units and scale
 - instrument/procedure used
 - timing (when measured)
 - quality control (calibration, inter-rater reliability, missingness)
-

Methods: missing data (plan + handling)

State:

- what “missing” means in your context
 - how much missingness occurred (Results often shows amounts)
 - handling strategy (complete case, imputation, model-based)
-

Methods: sample size and power (when relevant)

If confirmatory:

- target effect size / minimal important difference
- assumed variability or event rate
- alpha and power targets
- expected attrition

If exploratory, say so; don't pretend it was powered for everything.

Methods: statistical analysis plan (SAP) essentials

Readers need to know:

- primary outcome + primary comparison/model
 - covariates (pre-specified vs exploratory)
 - assumptions checked and diagnostics used
 - multiple comparisons adjustments (if relevant)
 - software (and version) used
-

Methods: report the analysis in reproducible units

At minimum:

- software + versions
 - packages/libraries + versions
 - random seeds (where used)
 - data processing steps (what changed, when, why)
-

Methods: preregistration and protocols (if any)

If preregistered:

- link to registration/protocol
- deviations and why (Discussion often revisits this)

If not preregistered, come clean.

Methods: ethics and transparency

State (as applicable):

- ethics approval body + reference
 - consent process
 - data privacy and anonymisation
 - conflicts of interest and funding
-

Methods: quick checklist

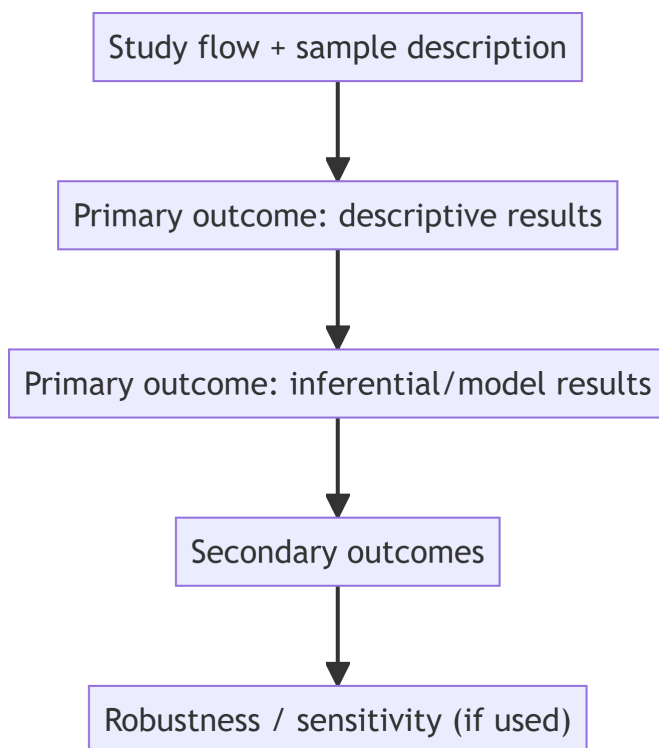
- Can a reviewer reproduce the analysis from your description?
 - Is the experimental unit unambiguous?
 - Are primary vs secondary outcomes distinct?
 - Are measurement units and timing explicit?
 - Is the SAP clear enough to prevent “analysis drift”?
-

Results (the “what answer did you get?” section)

Results should be:

- **evidence-first**
 - **structured**
 - **interpretation-light** (save meaning for Discussion)
-

Results: the story arc



Results: a simple discipline

For every key result, report:

- the **estimate**

- the **uncertainty** (CI/SE/SD depending on purpose)
- the **n** (denominator matters)
- and, if hypothesis testing is used, the **p-value**

Avoid reporting p-values without effect sizes.

Descriptive statistics: what they are for

Descriptives answer:

- “What does the data look like?”
- “Are values plausible?”
- “How variable is this phenomenon?”

Descriptives are not “inferior”; they are the foundation of trust.

Descriptive stats: choose summaries that match distributions

Common pairings:

- approximately symmetric: **mean $\pm$ SD**
- skewed / heavy tails: **median + IQR**
- counts: **n (%)**

Always include **units** and **time window** where relevant.

Descriptive stats: show distributions, not only summaries

Prefer visuals that reveal shape:

- histograms / density plots
- boxplots / violin plots
- jittered points (when n is moderate)

This prevents “summary masking” (different distributions, same mean).

Inferential statistics: what they are for

Inferentials answer:

- “Is the observed pattern plausibly due to chance under a null model?”
- “How big might the effect be in the population?”

Inferential work should align to the **analysis plan** stated in Methods.

Inferentials: what to report (minimum)

For each key test/model:

- effect estimate (difference, ratio, slope, etc.)
 - uncertainty (95% CI when possible)
 - test statistic (t, χ^2 , F, z) and degrees of freedom (when applicable)
 - p-value (with sensible precision)
-

Inferentials: practical vs statistical significance

Teach your reader the difference:

- statistical: unlikely under a null model
- practical: big enough to matter in the real world

Use domain thresholds when possible (clinically meaningful, policy meaningful, etc.).

Multiple comparisons (when it matters)

If you test many outcomes/groups:

- state how many tests were run
 - indicate whether adjustment was applied (and which)
 - consider prioritising:
 - one primary outcome
 - a small set of planned contrasts
-

Tables: why they still matter in 2026

Tables are best when readers need:

- exact values (not approximate)
- many numbers with consistent meaning
- compact comparison across groups/conditions

Figures are best for pattern recognition.

Table design principle: make tables scannable

Good tables are:

- **minimalist** (avoid heavy gridlines)
 - aligned (decimals align; text aligns)
 - explicit about denominators (n)
 - explicit about units
 - clear about what summary is shown (mean $\pm$ SD vs median [IQR])
-

Boxed vs open tables (a practical guideline)

- **Boxed tables:** strong outer border; useful for slides/handouts where the table floats visually.
- **Open/minimal tables:** no vertical lines; few horizontal rules; best for journals and readability.

Rule: reduce ink that doesn't carry data.

Table: a “minimalist” layout example (template)

Outcome	Group A (n=)	Group B (n=)	Effect (A–B)	95% CI	p
Primary outcome (units)	mean (SD)	mean (SD)	Δ	[L, U]	
Secondary outcome					

Notes (footnotes): definition of outcomes, time window, model adjustment, missingness handling.

Tables: common mistakes

- mixing different summaries in one column without labels
- unclear denominators in fractional values (n changes across rows but not indicated)
- too many decimal places (false precision)
- p-values without effect sizes
- “significant” stars without explanation

Figures: when to use a plot instead of a table

Use figures when the goal is:

- show distributions or variability
 - show relationships ($x \rightarrow y$)
 - compare patterns across many groups (small multiples)
 - communicate uncertainty visually (error bars, ribbons, intervals)
-

Visualisation: why design principles belong in Results

Results often fail because:

- graphs are decorative, not explanatory
- legends/axes hide the meaning
- comparisons are hard (no common scale, clutter)

Design is not cosmetic; it's evidential clarity.

Visualisation foundations (three lenses)

- **Perception:** how humans group and interpret marks
 - **Integrity:** graphical honesty + data-ink discipline
 - **Structure (Grammar of Graphics):** build plots as compositional statements
-

Gestalt principles (applied to Results figures)

Use intentionally:

- **Proximity:** group related items so that they are easy to spot
 - **Similarity:** use colour/shape consistently for meaning
 - **Figure-ground:** make data pop; de-emphasise scaffolding
 - **Focal point:** highlight what matters (sparingly)
-

The Grammar of Graphics (the mental model)

Think of a plot as a structured sentence:

- **DATA**: what you have
 - **AESTHETIC MAPPINGS** : how variables map to position/colour/size
 - **GEOMETRY**: points/lines/bars (what gets drawn)
 - **SCALES**: how values become colours, sizes, axes
 - **COORDS**: coordinate system
 - **GUIDES**: legends, labels, theme
-

A quick grammar table

Grammar component	Meaning	ggplot2 analogue
DATA	dataset	<code>data = ...</code>
TRANS	transformations	<code>log()</code> , <code>cut()</code> , <code>stat_*()</code>
SCALE	map values → aesthetics	<code>scale_*()</code>
COORD	coordinate system	<code>coord_*()</code>
ELEMENT	marks to draw	<code>geom_*()</code>
GUIDE	interpretation aids	<code>labs()</code> , <code>theme()</code> , <code>guides()</code>

Progressive construction of a ggplot (the discipline)

Build plots piece by piece:

- 1) empty canvas
- 2) add data
- 3) map aesthetics
- 4) add geometry
- 5) refine scales

- 6) add guides (labels/themes)
 - 7) facet (small multiples)
-

Example: start with an empty plot

```
library(ggplot2)
ggplot()
```

Meaning: empty coordinate space—no data, no mapping, no marks.

Example: add data

```
ggplot(data = iris)
```

Still nothing visible: we have not mapped variables or chosen a geometry.

Example: add aes() (mapping)

```
ggplot(
  data = iris,
  aes(x = Sepal.Length, y = Petal.Length)
)
```

Now we've specified *meaning* (what x and y represent), but still no marks.

Example: add a geometry (marks)

```
ggplot(iris, aes(Sepal.Length, Petal.Length)) +  
  geom_point()
```

This is the minimal “sentence”: plot y against x as points.

Example: scale and guide (make it readable)

```
ggplot(iris, aes(Sepal.Length, Petal.Length, colour = Species)) +  
  geom_point(size = 2) +  
  labs(  
    title = "Iris morphology",  
    x = "Sepal length (cm)",  
    y = "Petal length (cm)",  
    colour = "Species"  
  ) +  
  theme_minimal()
```

Example: facets (small multiples)

```
ggplot(iris, aes(Sepal.Length, Petal.Length, colour = Species)) +  
  geom_point(size = 2, alpha = 0.8) +  
  facet_wrap(~ Species) +  
  labs(title = "Iris morphology by species") +  
  theme_bw()
```

Facets often beat colour alone when the goal is comparison.

Uncertainty in plots: don't hide it

Prefer:

- intervals (CI ribbons, error bars) with clear definition
- raw points (when feasible) alongside summaries
- annotation that states what the interval means

Never assume the reader knows whether bars are SD/SE/CI.

Error bars: label them explicitly

In captions or notes, state:

- what the centre is (mean/median/model estimate)
 - what the bars represent (SD/SE/95% CI)
 - what n is (per group/condition)
-

Captions: a Results figure is incomplete without one

A good caption answers:

- what's plotted (variables + units)
 - what each colour/shape/facet means
 - what summary and uncertainty are shown
 - what model (if any) generated the estimate
 - any key data exclusions or transformations
-

Results writing: keep interpretation out

In Results, prefer:

- “Group A had higher X than Group B ($\Delta=...$, 95% CI ...)”

Avoid:

- “This demonstrates that...” (interpretation)
- “Surprisingly...” (interpretation)
- “Importantly...” (evaluation)

Save meaning for Discussion.

Results: a paragraph template (repeat for each key outcome)

- 1) **What** outcome and groups/conditions
 - 2) **Descriptive** summary (n, centre, spread)
 - 3) **Inferential** result (estimate, CI, p; model if relevant)
 - 4) **Pointer** to table/figure
-

Discussion (the “what does it mean?” section)

The Discussion:

- interprets results in context
 - explains plausibility and mechanisms
 - acknowledges uncertainty and limitations
 - avoids over-claiming
-

Discussion structure (practical)

- 1) one-paragraph **summary** of main findings
 - 2) **comparison** to prior literature (agreement/disagreement)
 - 3) **interpretation** (mechanisms, meaning)
 - 4) **limitations** (bias, confounding, measurement, generalisability)
 - 5) **implications** (practice/policy/theory)
 - 6) **future work** + closing take-home
-

Discussion: claims discipline

Match claim strength to design:

- RCTs → stronger causal claims (still bounded by context)
 - observational studies → cautious causal language; emphasise association + robustness
 - exploratory analyses → framed as hypothesis-generating
-

Discussion: limitations (write them like a scientist)

Good limitations:

- are specific (not generic “small sample size” only)
 - describe direction and impact (how might bias change estimates?)
 - show what you did to mitigate (sensitivity, robustness, measurement checks)
-

Discussion: avoid “spin”

Red flags:

- focusing only on significant results
 - changing the primary outcome after seeing results
 - overusing causal language when design doesn’t support it
 - burying null results
-

Conclusion slide (what you should remember)

- IMRaD is a disciplined way to answer a reader's four questions.
- Methods earn trust by making evaluation and replication possible.
- Results are evidence-first: **effect sizes + uncertainty**.
- Tables and figures are part of reporting integrity, not decoration.
- Discussion interprets with humility: claims proportional to evidence.